

Lesson 7: L'Hopital's Rule

1. Verification of Indeterminate Form (25%)

- **Criterion:** The student must explicitly show that direct substitution results in an indeterminate form, usually $0/0$ or ∞/∞ .
- **Evidence:** The student should write something similar to: " $\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = 0$ ".
- **Common Error:** Failing to evaluate the numerator and denominator separately, or failing to show the resulting form, can lose this point.

2. Application of L'Hopital's Rule (25%)

- **Criterion:** Correct differentiation of both the numerator and denominator separately.
- **Evidence:** The student clearly indicates LHR is used (usually by writing $= \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$) and correctly computes the derivatives.
- **Common Error:** Applying the Quotient Rule instead of differentiating the numerator and denominator separately results in 0 points for this step.

3. Correct Limit Evaluation (50%)

- **Criterion:** The resulting limit is correct, and notation is maintained.
- **Note:** If the limit is still indeterminate, LHR must be applied again.